

Skills Summary

Precise Definitions and Counterexamples

Use this reference while completing your worksheet or when studying.

1. Turn a definition into a coordinate test (test-a-definition-with-coordinates)

A definition you can compute with is a definition you can check.

congruent segments equal length: $AB = \sqrt{(\Delta x)^2 + (\Delta y)^2}$

midpoint of $\overline{AB}$ $M = \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$ — the one point with $AM = MB$

parallel slopes equal: $m_1 = m_2$

perpendicular slopes multiply to -1 : $m_1 m_2 = -1$

Read the definition, name the quantity it talks about, compute that quantity.

Example: Is $\triangle PQR$ isosceles? $P(0, 0)$, $Q(6, 0)$, $R(3, 4)$.

Step 1. “Isosceles” is defined by *at least two congruent sides*, and congruent means equal length — so this is a distance-formula question, not a looks-like-it question.

Step 2. $PR = \sqrt{(3-0)^2 + (4-0)^2} = \sqrt{9+16} = PR = 5$

Step 3. $QR = \sqrt{(3-6)^2 + 4^2} = 5$ too, and $PQ = 6$: two sides match, so yes, isosceles.

Try it: $\triangle ABC$ has $A(-5, 0)$, $B(5, 0)$, $C(0, 12)$. Compute AC to test whether it is isosceles.

check: $AC = 13$

2. Kill a claim with one counterexample (find-a-counterexample)

A counterexample to “if P then Q ” is one object where P is true and Q is false. One is enough — you never need a second, and you never need to explain why the claim “usually” fails.

Where to hunt: the claim names one property; look for an object with that property and nothing else. Equal *lengths* say nothing about *direction*; equal *angles* say nothing about *size*.

Converse warning: “if P then Q ” being true tells you nothing about “if Q then P ”. Each needs its own proof or its own counterexample.

Example: Disprove: “If two segments have the same length, they are parallel.” $\overline{AB}$: $(0, 0)$ to $(6, 8)$, length 10, slope $\frac{4}{3}$. $\overline{CD}$: $(0, 0)$ to $(8, 6)$, length 10.

Step 1. Both segments already satisfy the hypothesis (equal length), so all that is left is to show the conclusion fails — compare slopes.

Step 2. slope of $\overline{CD} = \frac{6-0}{8-0} = \frac{3}{4}$

Step 3. $\frac{3}{4} \neq \frac{4}{3}$, so they are not parallel. One object, claim dead.

Try it: $\overline{EF}$: (0, 0) to (5, 12) and $\overline{GH}$: (0, 0) to (12, 5) both have length 13. Compute the slope of $\overline{GH}$.

check: $\frac{5}{12}$

3. Repair a loose definition (write-a-precise-definition)

A definition must do two jobs: (1) be true of every object it names, and (2) be true of *nothing else*. A statement that only does job (1) is a **property** (a theorem), not a definition.

Repair recipe: name the general category, then add exactly the conditions that rule everything else out — and test the repair by hunting for one more counterexample.

Example: “The perpendicular bisector of $\overline{AB}$ is a line through the midpoint of $\overline{AB}$.” Test it with $A(0, 0)$, $B(8, 4)$, midpoint $M(4, 2)$, and the line through M and $P(6, 5)$.

Step 1. To show the wording admits too much, find one line that fits it and is not the perpendicular bisector — so compute the candidate’s slope.

Step 2. slope of $\overrightarrow{MP} = \frac{5 - 2}{6 - 4} = \frac{3}{2}$

Step 3. $\overline{AB}$ has slope $\frac{1}{2}$, so perpendicular would need -2 . The line goes through M but is not perpendicular: the wording fails job (2). **Repair:** the line through the midpoint of $\overline{AB}$ and *perpendicular to $\overline{AB}$* .

Try it: “A median goes from a vertex to the opposite side” is too loose. In $\triangle ABC$ with $A(-2, 0)$, $B(8, 0)$, $C(1, 7)$, find the midpoint of $\overline{AB}$ — the endpoint the real median must hit.

check: (3, 0)

(!) Watch out: “Not a definition” does not mean “false”. Every rhombus really does have perpendicular diagonals — that statement is a true property. It fails only as a *definition*, because kites have perpendicular diagonals too.